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Secure Packages with CodeArtifact

yakubu abdullahi yusuf

Package name	Namespace	Format	Latest version	Latest publish date	Publish	Upstream
backport-util-concurrent	backport-util-concurrent	maven	3.1	1 minute ago	Block	Allow
classworlds	classworlds	maven	1.1	1 minute ago	Block	Allow
google	com.google	maven	1	1 minute ago	Block	Allow
jsr305	com.google.code.findbugs	maven	2.0.1	1 minute ago	Block	Allow
google-collections	com.google.collections	maven	1.0	1 minute ago	Block	Allow
commons-cli	commons-cli	maven	1.0	1 minute ago	Block	Allow
commons-logging-api	commons-logging	maven	1.1	1 minute ago	Block	Allow

Introducing Today's Project!

In this project, I will demonstrate how to set up and use AWS CodeArtifact to manage and secure Java dependencies for my applications. I'm doing this project to learn CICD and DevOps at large.

Key tools and concepts

Services I used were EC2 instance, Maven, CodeArtifact and ArtifactRepository. Key concepts I learnt include setting up and configuring AWS CodeArtifact, Use IAM roles and policies to let my EC2 instance access CodeArtifact.

Project reflection

This project took me approximately 40 minutes and the most rewarding about it is learning and practicalising new things.

This project is part three of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow so stay tuned

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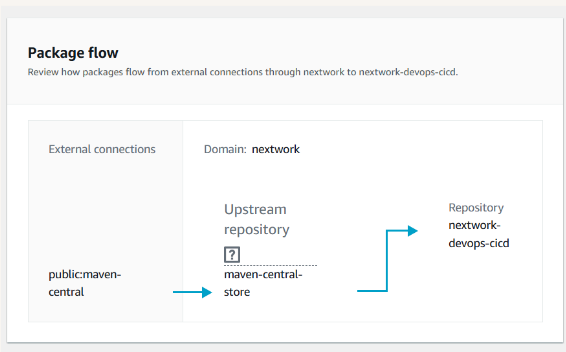
CodeArtifact Repository

CodeArtifact is a secure and central place for storing all software packages. Engineering teams use artifact repositories because it is secure, reliable and easy to access and share packages with each other.

A domain is a folder that keeps multiple repositories from the same organization or projects together. It helps in setting up permissions and security settings that apply to all repositories at once rather than seperately. My domain is nextwork.

A CodeArtifact repository can have an upstream repository, which acts as external sources that your main repository can pull from when a required package or dependency isn’t available. My repository's upstream repository is MAven central.



CodeArtifact Security

Issue

To access CodeArtifact, we need an authorizaton token because it ensures that only authorized systems can interact with your private package repositories. I ran into an error when retrieving a token because my ec2 isnt authorized to access repository

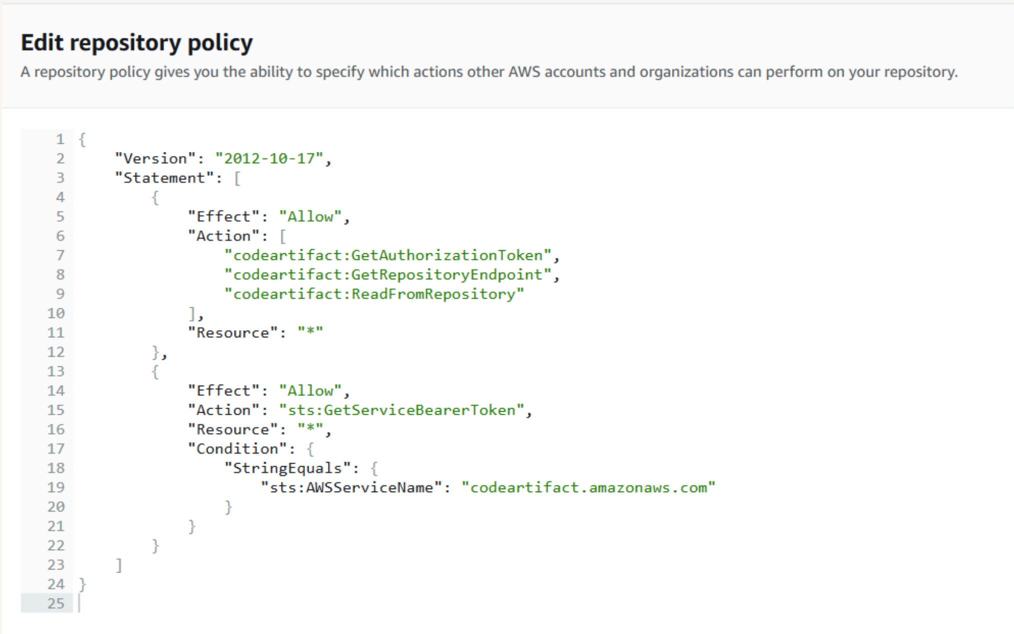
Resolution

To resolve the error with my security token, I created IAM policy, assigned it to a role then finally assign the role to my EC2 instance. This resolved the error because it gave my EC2 permissions to have access to the repository.

It's security best practice to use IAM roles because AWS automatically issues and refreshes temporary security credentials for the instance.

The JSON policy attached to my role

The JSON policy I set up grants access to security tokens, read packages from repositories and repository location. It also allows elevated access specifically for CodeArtifact operations temporarily.



Edit repository policy

A repository policy gives you the ability to specify which actions other AWS accounts and organizations can perform on your repository.

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "codeartifact:GetAuthorizationToken",
                "codeartifact:GetRepositoryEndpoint",
                "codeartifact:ReadFromRepository"
            ],
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": "sts:GetServiceBearerToken",
            "Resource": "*",
            "Condition": {
                "StringEquals": {
                    "sts:AWSServiceName": "codeartifact.amazonaws.com"
                }
            }
        }
    ]
}
```

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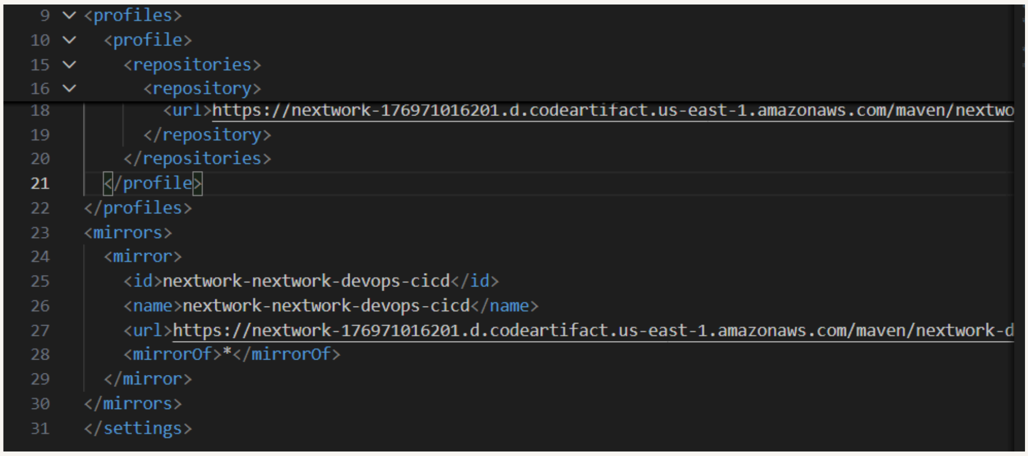
Maven and CodeArtifact

To test the connection between Maven and CodeArtifact, I compiled my web app using settings.xml

The settings.xml file configures Maven to connect to repositories and handle authentication. it tells Maven where to find your CodeArtifact repository and how to securely authenticate with it.

Compiling means converting myvproject's code into a format the computer can understand and execute. It ensures everything is correctly configured and ready to become a working application.



```
<profiles>
  <profile>
    <repositories>
      <repository>
        <url>https://nextwork-176971016201.d.codeartifact.us-east-1.amazonaws.com/maven/nextwo
      </repository>
    </repositories>
  </profile>
</profiles>
<mirrors>
  <mirror>
    <id>nextwork-nextwork-devops-cicd</id>
    <name>nextwork-nextwork-devops-cicd</name>
    <url>https://nextwork-176971016201.d.codeartifact.us-east-1.amazonaws.com/maven/nextwork-d
    <mirrorOf>*</mirrorOf>
  </mirror>
</mirrors>
</settings>
```

Verify Connection

After compiling, I checked my Artifact Repositoty and I noticed the repository being populated by packages.

	Package name	Namespace	Format	Latest version	Latest publish date	Publish	Upstream
○	backport-util-concurrent	backport-util-concurrent	maven	3.1	1 minute ago	Block	Allow
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○	google	com.google	maven	1	1 minute ago	Block	Allow
○	jsr305	com.google.code.findbugs	maven	2.0.1	1 minute ago	Block	Allow
○	google-collections	com.google.collections	maven	1.0	1 minute ago	Block	Allow
○	commons-cli	commons-cli	maven	1.0	1 minute ago	Block	Allow
○	commons-logging-api	commons-logging	maven	1.1	1 minute ago	Block	Allow

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